

GATE 3 — HEATER CIRCUIT VERIFICATION RECORD

CIP-BTSB-BT-001 · Supporting Record G3

CIP — Compute Infrastructure Platform · S. Mohammed Idreesh · Chennai Test date: 25 August 2026 · Instrument: in-line DC voltmeter/ammeter module

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

To verify, before any heat is applied to the test fixture, that the heater circuit:

- 1. is wired in **series** and not in parallel — the failure mode that would deliver nine times the intended power and destroy the heaters;
- 2. delivers a power within the **10-15 W band** pre-registered in Section 5 of the test report;
- 3. delivers that power **evenly across all three heaters**;
- 4. can be **measured**, so that the report can state delivered power as a measured quantity rather than a derived one.

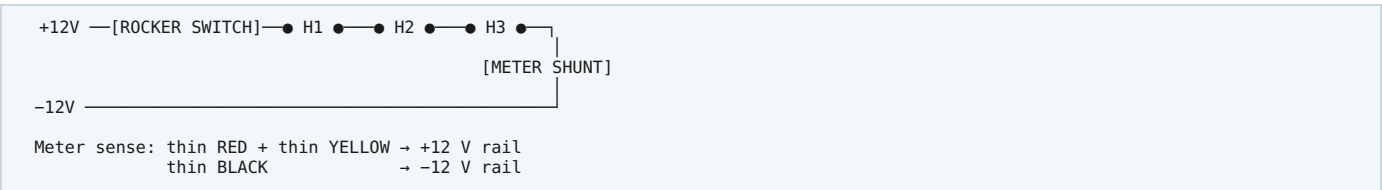
This record is referenced by Sections 4 and 5 of the pre-registered test report CIP-BTSB-BT-001.

2. CONFIGURATION AS BUILT

2.1 Components

Item	Detail
Supply	SEM-1250-L2 AC/DC adapter, 100-260 V AC in, 12 V DC / 5.0 A max out, centre-positive, Made in India
Supply termination	Female DC barrel jack, screw terminals, red = +12 V confirmed
Heaters	3 × ceramic cartridge heater, 6 × 20 mm, 12 V 40 W , red fibreglass-braided silicone leads
Switch	SPST rocker switch, in the positive line, serving as manual abort
Instrument	Dual-display DC voltmeter/ammeter module, internal shunt, 3-pin sense connector + 2-pin current connector
Interconnect	18 AWG silicone (current path), PVC hook-up wire (switch link)

2.2 Circuit topology



The shunt is placed in the **low side** (return path), so that it shares the ground reference of the meter’s sense input. Three heaters share a single series loop; the loop has exactly two free ends, verified by physical count and by continuous finger trace before power was applied.

2.3 Deviation from the build manual

The LM2596 buck converter module was removed from the design and is not fitted.

Rationale: the supply already delivers 12 V, which is the heaters’ design voltage. An LM2596 requires approximately 1.5 V of headroom and would have reduced delivered power to about 10 W. It would also have introduced an adjustable potentiometer whose accidental movement between runs would silently destroy run-to-run comparability. Its removal reduces the component count and eliminates a failure mode.

3. METHOD

Two measurements were taken.

Test 1 — no-load. The meter’s three sense wires were connected across the supply with both current-path conductors left open and insulated. This confirms supply voltage, supply polarity, meter function, and that the shunt is genuinely isolated.

Test 2 — five-second loaded. The complete series loop was closed, the three heaters resting free on a non-combustible surface, not in contact with each other or with the fixture. Power was applied for five seconds and voltage and current read during that interval. Exposure was deliberately limited to five seconds so that a mis-wired parallel circuit would be interrupted before thermal damage.

A tactile check of all three heater bodies followed immediately.

4. RESULTS

4.1 Test 1 — no-load

Indicated voltage	12.0 V
Indicated current	0.00 A
Supply polarity	Centre-positive confirmed; red = +12 V

0.00 A is the correct and expected reading: with the current-path conductors open, no current traverses the shunt. A non-zero reading here would have indicated an unintended path.

4.2 Test 2 — five-second loaded

Indicated voltage	11.9 V
Indicated current	1.19 A
Exposure	5 s

4.3 Derived quantities — the governing values

DELIVERED ELECTRICAL POWER			
$P = V \times I = 11.9 \times 1.19$	=	14.16 W	→ 14.2 W
TOTAL LOOP RESISTANCE			
$R = V \div I = 11.9 \div 1.19$	=	10.00 Ω	
PER-HEATER RESISTANCE			
$R_1 = 10.00 \div 3$	=	3.33 Ω	
PER-HEATER POWER			
$P_1 = 14.16 \div 3$	=	4.72 W	
BARREL HEAT FLUX			
$A = \pi \times 0.6 \text{ cm} \times 2.0 \text{ cm}$	=	3.77 cm²	
$q = 4.72 \div 3.77$	=	1.25 W/cm²	

4.4 Series topology — confirmation by resistance

This is the governing safety result.

Wiring	Predicted R	Predicted I at 12 V	Observed
Three in series	10.0 Ω	1.20 A	✓ 1.19 A
Two in series	6.67 Ω	1.80 A	—
One heater only	3.33 Ω	3.60 A	—
Three in parallel	1.11 Ω	10.8 A (supply-limited)	—

Measured loop resistance of 10.00 Ω is a factor of nine above the parallel case and matches the series prediction to within 1%. Series topology is confirmed by measurement, independently of the physical count.

Nominal per-heater resistance from the 12 V / 40 W rating is 3.60 Ω; measured is 3.33 Ω, i.e. **7.4% below nominal**. This is within ordinary manufacturing tolerance for ceramic cartridge heaters and is the reason delivered power is 14.2 W rather than the 13.3 W predicted from nameplate values.

4.5 Thermal distribution check

All three heater bodies were touched immediately after the five-second exposure.

Heater	Result
H1	Warm
H2	Warm
H3	Warm

All three warm, none hot, no odour, no discolouration. An open joint would have left one heater cold; a parallel fault would have made all three too hot to touch within the same interval.

5. INTERPRETATION

5.1 Against the pre-registered power band

Section 5 of CIP-BTSB-BT-001 pre-registers a delivered-power band of **10-15 W**.

Measured 14.2 W falls inside that band. No change to the pre-registered analysis plan, abort thresholds or predicted intervals is required.

5.2 Expected plate response

Using the nominal plate pair (2 × 99 g of 5052 aluminium, $c = 0.88 \text{ J/g}\cdot\text{K}$):

Hot plate alone	87.1 J/K	→	14.16 / 87.1 = 0.163 K/s	=	9.8 °C/min
Plate pair	174.2 J/K	→	14.16 / 174.2 = 0.081 K/s	=	4.9 °C/min

Expected TC1 rise in the first minute of Run C: 5-10 °C. A rise substantially above this band during a run indicates a wiring fault that developed after this record was taken, and is grounds for abort.

These figures use nominal plate mass. They must be recomputed once both plates are weighed.

5.3 Expected drift during a run

Metallic resistance rises with temperature. Current is therefore expected to fall slightly as the heaters warm — approximately **1.15 A and 13.7 W** at steady state. This is normal behaviour and is not an abort condition. Voltage and current are to be logged at the start and end of every run.

5.4 Instrument caveat

The meter is an uncalibrated commercial module of unstated accuracy, typically $\pm 1\% \pm 2$ digits. The 14.2 W figure should therefore be read as **14.2 ± 0.5 W**.

It is not the primary power measurement. **Run C calibrates delivered power thermally**, from the plate pair’s known heat capacity and the observed rate of temperature rise, and that thermal figure governs the energy balance in Section 6 of the test report. The electrical measurement in this record is a cross-check and a safety verification, not the input to the analysis.

6. CORRECTION TO EARLIER FAULT-DIAGNOSIS GUIDANCE

Recorded because it is part of the instrument’s history and because the erroneous values were in the operator’s hands during the test.

Error. The field diagnostic table supplied to the operator before Test 2 listed “0.3-0.4 A = one heater only”, “0.6-0.8 A = two heaters only”, and “3.0-3.5 A = parallel”. **All three figures were wrong.**

Correct values. In a series loop an open joint yields 0.00 A, not a reduced current — there is no partial-conduction case. A single heater across 12 V draws 3.6 A, not 0.35 A. Three heaters in parallel demand 10.8 A, which exceeds the supply’s 5 A rating and would present as collapsing voltage rather than as a stable 3.3 A reading.

Consequence. None to this test. The observed 1.19 A matched the correct series prediction, and the independent physical checks — free-end count, finger trace, and tactile check — were valid and were passed. The corrected table in Section 4.4 supersedes the earlier guidance and is the version to be used for all subsequent runs.

7. OUTSTANDING BEFORE FIRST HEATED RUN

[]	Heater leads trimmed to 500 / 200 / 200 / 200 / 200 / 500 mm
[]	RE-MEASURE voltage and current after trimming (removing ~4.3 m from the series path lowers loop resistance by an estimated 0.15-0.20 Ω and will raise delivered power by approximately 3%. The trimmed configuration is the one that runs the test, so the trimmed configuration is the one that must be recorded.)
[]	Heaters bonded to plate H at 43 / 86 / 130 mm with thermal compound and Kapton restraint
[]	Both plates weighed; Section 5.2 recomputed with real masses
[]	Pre-registered report CIP-BTSB-BT-001 corrected to state three heaters in series and measured power, then timestamped BEFORE the first heated run

The re-measurement after trimming is not optional. This record documents the untrimmed configuration. Reporting an untrimmed power figure against a trimmed experiment would be a misstatement of method, and it is exactly the kind of discrepancy that destroys confidence in everything else in the report.

8. GATE STATUS

GATE 3 — PASSED for series topology, power delivery within the pre-registered band, even distribution across all three heaters, and instrument function.

Provisional pending re-measurement after lead trimming — Section 7.

Record G3 · Issued 25 August 2026 · Supersedes the fault-diagnosis table issued verbally prior to Test 2. Supports CIP-BTSB-BT-001. All build-manual performance figures remain [GEN-0 VALIDATION REQUIRED].